

# ⌈ Inequalities of Real Numbers.

## ⌈ Bernoulli's Inequality

If  $a > -1$ ,  $1 + na \leq (1+a)^n$  for all  $n \in \mathbb{N}$ .

Proof. If  $n=1$ , trivial. If the statement hold for  $n=k \in \mathbb{N}$ , then

$$(1+a)^{k+1} = (1+a)(1+a)^k \underset{1+a \geq 0}{\geq} (1+a)(1+ka) = 1 + (k+1)a + a^2 \geq 1 + (k+1)a.$$

## ⌈ Arithmetic-Geometric mean

If  $a, b \in \mathbb{R}$ , then  $2ab \leq a^2 + b^2$  (Directly, it can be proved by  $(a-b)^2 \geq 0$ .)

## ⌈ Generalized A-G mean inequality,

If  $a_i \geq 0$ , then for all  $n \in \mathbb{N}$ ,

$$(a_1 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} (a_1 + a_2 + \cdots + a_n)$$

proof. If  $n=2$ ,  $\sqrt{a_1 a_2} \leq \frac{1}{2}(a_1 + a_2)$ , thus the statement holds.

Suppose that for  $n=2^k$  the statement be hold. There is

$$(a_1 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} (a_1 + \cdots + a_n)$$

$$\begin{aligned} \text{Now, } \frac{1}{2n} (a_1 + \cdots + a_{2n}) &= \frac{1}{2} \left( \frac{1}{n} (a_1 + \cdots + a_n) + \frac{1}{n} (a_{n+1} + \cdots + a_{2n}) \right) \\ &\geq \frac{1}{2} \left( (a_1 \cdots a_n)^{\frac{1}{n}} + (a_{n+1} \cdots a_{2n})^{\frac{1}{n}} \right) \\ &= \frac{1}{2} \left( \left( (a_1 \cdots a_n)^{\frac{1}{2n}} \right)^2 + \left( (a_{n+1} \cdots a_{2n})^{\frac{1}{2n}} \right)^2 \right) \\ &\geq \frac{1}{2} \cdot 2 (a_1 \cdots a_n)^{\frac{1}{2n}} \cdot (a_{n+1} \cdots a_{2n})^{\frac{1}{2n}} \\ &= (a_1 \cdots a_n a_{n+1} \cdots a_{2n})^{\frac{1}{2n}} \end{aligned}$$

Therefore, by induction, the statement hold when  $n=2^k$  ( $k \in \mathbb{N}$ ).

To generalize, let  $n \geq 2$  be given put  $k \in \mathbb{N}$  s.t

$$2^{k-1} \leq n \leq 2^k \quad \frac{1}{2^k} \leq \frac{1}{n} \leq \frac{1}{2^{k-1}} \quad 1 \leq 2$$

put

$$b_i = \begin{cases} a_i & \text{if } i \leq n \\ \frac{1}{n} (a_1 + \cdots + a_n) = M & \text{if } n < i \leq 2^k \end{cases} \quad \frac{n}{2^k} \leq 1 \quad 0 \leq \left( -\frac{n}{2^k} \right)$$

$$\text{Then, } (b_1 \cdots b_{2^k})^{\frac{1}{2^k}} \leq \frac{1}{2^k} \cdot (b_1 + \cdots + b_{2^k}) = \frac{1}{2^k} \cdot (nM + (2^k - n)M) = M$$

$$\text{And } (b_1 \cdots b_{2^k})^{\frac{1}{2^k}} = (a_1 \cdots a_n \cdot M^{2^k - n})^{\frac{1}{2^k}} = (a_1 \cdots a_n)^{\frac{1}{2^k}} \cdot M^{(1 - \frac{n}{2^k})}$$

$$\text{Finally, } (a_1 \cdots a_n)^{\frac{1}{2^k}} \leq M^{\frac{n}{2^k}} \text{ implies } (a_1 \cdots a_n)^{\frac{1}{n}} \leq (a_1 + \cdots + a_n) \cdot \frac{1}{n}.$$

┌ Lagrange's Inequality.

Let  $a_i, b_i \in \mathbb{R}$ . Then,

$$\left[ \sum_{i=1}^n a_i b_i \right]^2 = \left[ \sum_{i=1}^n a_i^2 \right] \cdot \left[ \sum_{i=1}^n b_i^2 \right] - \frac{1}{2} \cdot \sum_{1 \leq j, k \leq n} (a_j b_k - a_k b_j)^2$$